

DEPARTMENT OF ELECTRONICS AND INSTRUMENTATION

NATIONAL LEVEL

AUTOMATION & ROBOTICS CHALLENGE - 2026

FOURTH EDITION

ARC'26



Showdown Event : 02 & 03 March 2026

Venue: Main Campus, SRM IST, Kattankulathur.

About SRM Institute of Science and Technology

SRM Institute of Science and Technology (formerly SRM University) is a top Indian university offering diverse UG, PG, and doctoral programs. Accredited with an 'A++' by NAAC and classified as a category-I University by the UGC and Ministry of Education, SRMIST ranks 4th among private institutions in ARIIA-2021. QS rates it as a "4 Star" institute with "5 Star" ratings in Teaching, Employability, Inclusiveness, and Online Teaching. SRMIST has been ranked 11th nationally under Universities Category by NIRF (National Institutional Ranking Framework) in 2025.

About the Department

The Department of Electronics & Instrumentation Engineering was established in 2007-08, offering B.Tech degrees in Electronics & Instrumentation Engineering and Automation and Robotics, as well as M.Tech degrees in Electronics and Control and Automation and Robotics. The department's programs cater to the expanding needs of manufacturing and process industries. Faculty and students have filed over 100+ patents, with 75 granted and over 100 published. The department has secured Rs. 2 crores in funding from DST, SERB, SPARC, and NewGen IEDC. Accredited by IET and recognized by the IASC Sector Skill Council (affiliated with the Ministry of Skill Development and Entrepreneurship, Govt. of India), it also established a Center of Excellence in Factory Automation in collaboration with Mitsubishi Electric India Pvt. Ltd. and Center of Excellence in HMI in collaboration with Schneider Electric India Pvt. Ltd.

Introduction

The National Level Automation & Robotics Challenge 2026, hosted by the Department of Electronics and Instrumentation Engineering at SRM Institute of Science and Technology, is organized by the SRM Automation Club. This event is designed to engage college students with the evolving landscape of Automation & Robotics, where the students will address pressing challenges and devise innovative solutions in our 4 Theme Realms. This provides a valuable exposure to real-world applications and emerging innovations, encouraging participants to deepen their knowledge and explore potential advancements in the AI-Driven World!



Rules and Guidelines

- Participants are advised to stick to the deadlines and abide by the rules.
- A team can have a minimum of **2 participants** and maximum of **4 participants**.
- A team should consist of members representing the **same College/University**.
- Each team is restricted to choose only **a single theme out of four**.
- Participants are not allowed to represent **two teams** even with different themes. Participants are instructed to fill appropriate details in the Google registration form.
- All teams must assemble their project during the challenge event itself. **Pre-assembled or fully built setups** brought from outside are strictly not allowed.
- Participants must **bring all necessary components, tools, and materials required** for assembling their project. The organizers will not provide any build materials.

| Themes

1.Automation in Daily life

Smart Home Energy Management System:

Design an automated system that intelligently controls lights, fans, and air-conditioning based on occupancy, ambient light, and temperature to minimize energy consumption without compromising comfort.

Automated Parking Management System:

Develop an automated parking management system that detects slot availability, guides vehicles, and maintains usage data to reduce congestion and idle time.

Automated Domestic Water Management System:

Develop an automated parking management system that detects slot availability, guides vehicles, and maintains usage data to reduce congestion and idle time.

2.Industrial Automation

Automated Conveyor Sorting System:

Design an automated conveyor system that uses sensors and a PLC or microcontroller to identify objects by size, colour, or material and sort them using pneumatic diverters or servo motors.

PLC-based Automated Batch Process:

Design a PLC-based automated batch process for mixing, heating, and filling using sequential control, with level, flow, and temperature sensors for monitoring and an HMI for recipe management and operator interaction.

Machine Vision Defect Detection System:

Design a machine vision system using a camera with OpenCV or an industrial vision sensor, integrated with a PLC for accept/reject decisions and statistical quality reporting for process monitoring.

3. Agricultural Robotics

Targeted Crop Spraying Robot:

Design and develop a mobile robot that identifies crop zones and sprays only targeted areas, avoiding healthy regions.

Weed Detection and Removal Robot:

Develop a weed detection and removal robot that identifies weeds using sensors or vision systems. The robot should mechanically remove or push the weeds aside to improve agricultural efficiency.

Robotic Seed Sowing System:

Develop a robotic seed sowing system that moves autonomously and dispenses seeds at precise, controlled intervals for uniform planting.

4. Robotics in Healthcare

Adaptive Prosthetic Gripping Mechanism:

Design an adaptive gripping mechanism for a prosthetic arm that detects object shapes and dimensions using sensors. Feedback-based control should adjust grip force and finger position to ensure safe and stable grasping.

Vision-based Surgical Tool Localization:

Design a vision-based system to localize surgical tools during laser ablation surgery using real-time camera images. Image processing algorithms should track tool position and orientation to enhance precision and safety.

Automated Biomedical Waste Segregation System:

Design an automated biomedical waste segregation system to identify and classify different types of medical waste using sensors and vision techniques. The system processes this data to ensure safe, efficient, and hygienic waste segregation.

| Evaluation

Showdown Round

Participants are required to bring their own components & tools for developing their prototype. Each team can work in the designated workstations assigned to them. The Jury panel will evaluate the prototype based on the Evaluation Criteria to determine the winning teams.

Mode of Conduct: Offline

Date: 02-02-2026 to 03-02-2026

Venue: Main Campus, SRM IST ,Kattankulathur.

Showdown Entry Pass :

Registration Fee: Rs1500/-PER TEAM

What are all included in the Entry Pass?

- Lunch
- Refreshment
- Stationery kits
- Accolades & Certificates

Registration Link & QR :

<https://forms.gle/FkQ42Dz7asAYa8yi9>



Important Date :

Registration Phase : 28-01-2026 to 20-02-2026

Showdown Event : 02-03-2026 to 03-03-2026

CHIEF PATRONS

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Dr. P. Sathyanarayanan, Pro Chancellor (Academics), SRMIST

Dr. R. Shivakumar, Vice President, SRMIST

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Dr. A. Brindha, Assistant Professor, EIE

Dr. S. Sharanya, Assistant Professor, EIE

Dr. S. Indirani, Assistant Professor, EIE

Committee Leads

- 1.Vishnu Pradap D, President, Automation Club.
- 2.Mohan Sai, Technical Committee Lead.
- 3.Tarun Krishan, Registration Committee Lead.

Committee Members

Pranav | PriyanshSahu | Tharun | Moundy Raj | Shaswath Naga |
Brindha | Patric | Arijith | Ojeash | Monish Kanna | Darshan |
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